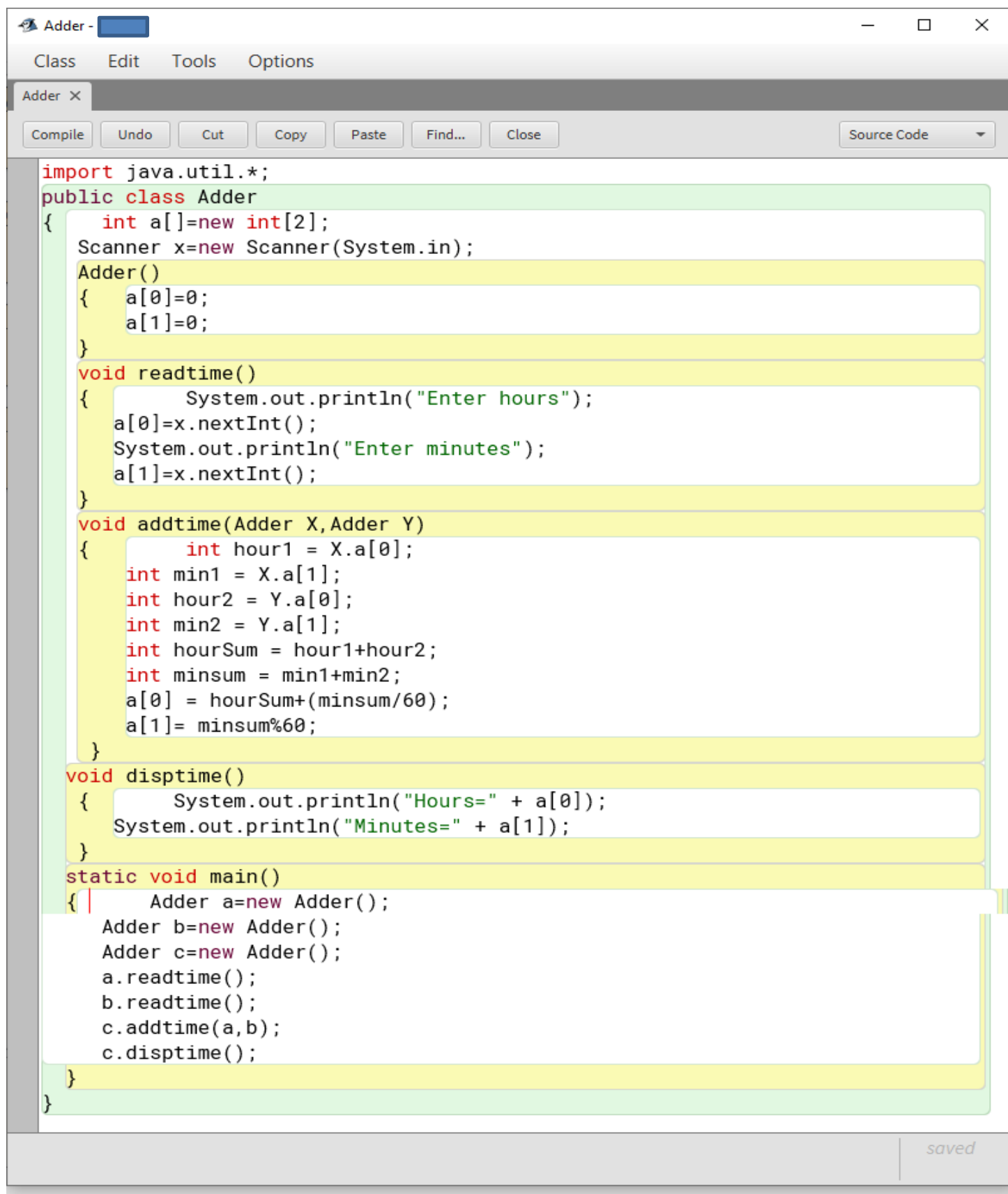
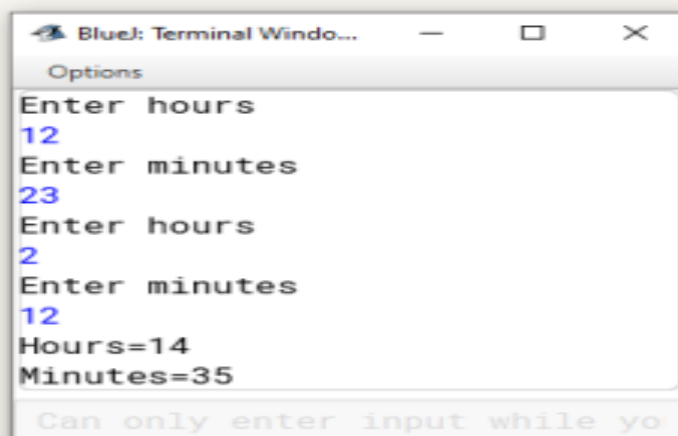




```
import java.util.*;
public class Adder
{
    int a[]=new int[2];
    Scanner x=new Scanner(System.in);
    Adder()
    {
        a[0]=0;
        a[1]=0;
    }
    void readtime()
    {
        System.out.println("Enter hours");
        a[0]=x.nextInt();
        System.out.println("Enter minutes");
        a[1]=x.nextInt();
    }
    void addtime(Adder X,Adder Y)
    {
        int hour1 = X.a[0];
        int min1 = X.a[1];
        int hour2 = Y.a[0];
        int min2 = Y.a[1];
        int hourSum = hour1+hour2;
        int minsum = min1+min2;
        a[0] = hourSum+(minsum/60);
        a[1]= minsum%60;
    }
    void disptime()
    {
        System.out.println("Hours=" + a[0]);
        System.out.println("Minutes=" + a[1]);
    }
    static void main()
    {
        Adder a=new Adder();
        Adder b=new Adder();
        Adder c=new Adder();
        a.readtime();
        b.readtime();
        c.addtime(a,b);
        c.disptime();
    }
}
```

saved



```
Blue: Terminal Windo...
Options
Enter hours
12
Enter minutes
23
Enter hours
2
Enter minutes
12
Hours=14
Minutes=35
Can only enter input while yo
```

BinSearch - [icon]

Class Edit Tools Options

BinSearch X

Compile Undo Cut Copy Paste Find... Close Source Code

```
import java.util.*;
class BinSearch
{
    int arr[];
    int n;
    static Scanner x=new Scanner(System.in);
    BinSearch(int nn)
    {
        n=nn;
    }
    void fillarray()
    {
        arr=new int[n];
        System.out.println("Enter "+n+ " elements");
        for(int i =0;i<n;i++)
            arr[i]=x.nextInt();
    }
    void sort()
    {
        int t;
        for(int i=0;i<n-1;i++)
            for(int j =0;j<n-1-i;j++)
            {
                if (arr[j]>arr[j+1])
                {
                    t=arr[j];
                    arr[j]=arr[j+1];
                    arr[j+1]=t;
                }
            }
    }
    int bin_search(int l,int u, int v )
    {
        int m=(l+u)/2;
        if(arr[m]==v)
            return m;
        else if(l>u)
            return -1;
        else if (arr[m]>v)
            return bin_search(l,m-1,v);
        else
            return bin_search(m+1,u,v);
    }
    static void main()
    {
        BinSearch obj = new BinSearch(5);
        obj.fillarray();
        obj.sort();
        System.out.println(" location: " + obj.bin_search(0,4,20) );
    }
}
```

saved

BlueJ:...

Options

Enter 5 elements

1  
2  
3  
4  
5

location: -1

Can only enter input while your

Capital - Class\_12

Class Edit Tools Options

Capital X

Compile Undo Cut Copy Paste Find... Close Source Code

```
import java.io.*;
import java.util. Scanner;
import java.util.StringTokenizer;
public class Capital{
private String sent;
private int freq;
public Capital() {
sent = new String();
freq = 0;
}
public void input() throws IOException {
Scanner sc = new Scanner(System.in);
System.out.print("Enter the sentence: ");
sent = sc.next();
}
boolean isCap(String w) {
char ch = w.charAt(0);
if(Character.isLetter(ch) && Character.isUpperCase(ch))
return true;
return false;
}
public void display() {
StringTokenizer st = new StringTokenizer(sent, " ");
int count = st.countTokens();
for (int i = 1; i<= count; i++) {
String word = st.nextToken();
if(isCap(word))
freq++;
}
System.out.println("Sentence: " + sent);
System.out.println("Frequency: " + freq);
}
public static void main(String args[ ])throws IOException {
Capital obj = new Capital();
obj.input();
obj.display();}
}
```

saved

BlueJ: Terminal Window - Class\_12

Options

Enter the sentence: Ram is a boy  
Sentence: Ram  
Frequency: 1

Can only enter input while your programming is

```
import java.util.Scanner;

public class DateValidator {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input the date
        System.out.print("Enter a date (YYYY-MM-DD): ");
        String dateString = scanner.nextLine();

        // Validate the date
        if (!isValidDate(dateString)) {
            System.out.println("Invalid date format or date out of range.");
            return;
        }

        // Calculate the future date
        String futureDate = calculateFutureDate(dateString, 100);

        // Print the future date
        System.out.println("Date after 100 days: " + futureDate);
    }

    // Validate date format and range
    public static boolean isValidDate(String dateString) {
        try {
            String[] parts = dateString.split("-");
            if (parts.length != 3) {
                return false;
            }

            if (year < 0 || month < 1 || month > 12 || day < 1 || day > 31) {
                return false;
            }

            // Check for valid number of days in the month
            if (month == 2 && day > 29) {
                return false;
            }
            if ((month == 4 || month == 6 || month == 9 || month == 11) && day > 30) {
                return false;
            }

            // Check for leap year and February
            if (month == 2 && day > 29 && !isLeapYear(year)) {
                return false;
            }

            return true;
        } catch (NumberFormatException e) {
            return false;
        }
    }

    // Calculate future date
    public static String calculateFutureDate(String dateString, int days) {
        try {
            String[] parts = dateString.split("-");
            int year = Integer.parseInt(parts[0]);
            int month = Integer.parseInt(parts[1]);
            int day = Integer.parseInt(parts[2]);

            // Convert the date to days since the beginning of the year
            int daysSinceStartOfYear = day;
            for (int i = 1; i < month; i++) {
                daysSinceStartOfYear += daysInMonth(year, i);
            }
        }
    }
}
```

```
// Add the number of days
daysSinceStartOfYear += days;

// Convert back to year, month, and day
while (daysSinceStartOfYear > daysInYear(year)) {
    daysSinceStartOfYear -= daysInYear(year);
    year++;
}

// Find the month
int currentMonth = 1;
while (daysSinceStartOfYear > daysInMonth(year, currentMonth)) {
    daysSinceStartOfYear -= daysInMonth(year, currentMonth);
    currentMonth++;
}

// Find the day
int currentDay = daysSinceStartOfYear;

// Format the date
String formattedDate = String.format("%04d-%02d-%02d", year, currentMonth, currentDay);
return formattedDate;
} catch (Exception e) {
    return "Error calculating future date";
}
}

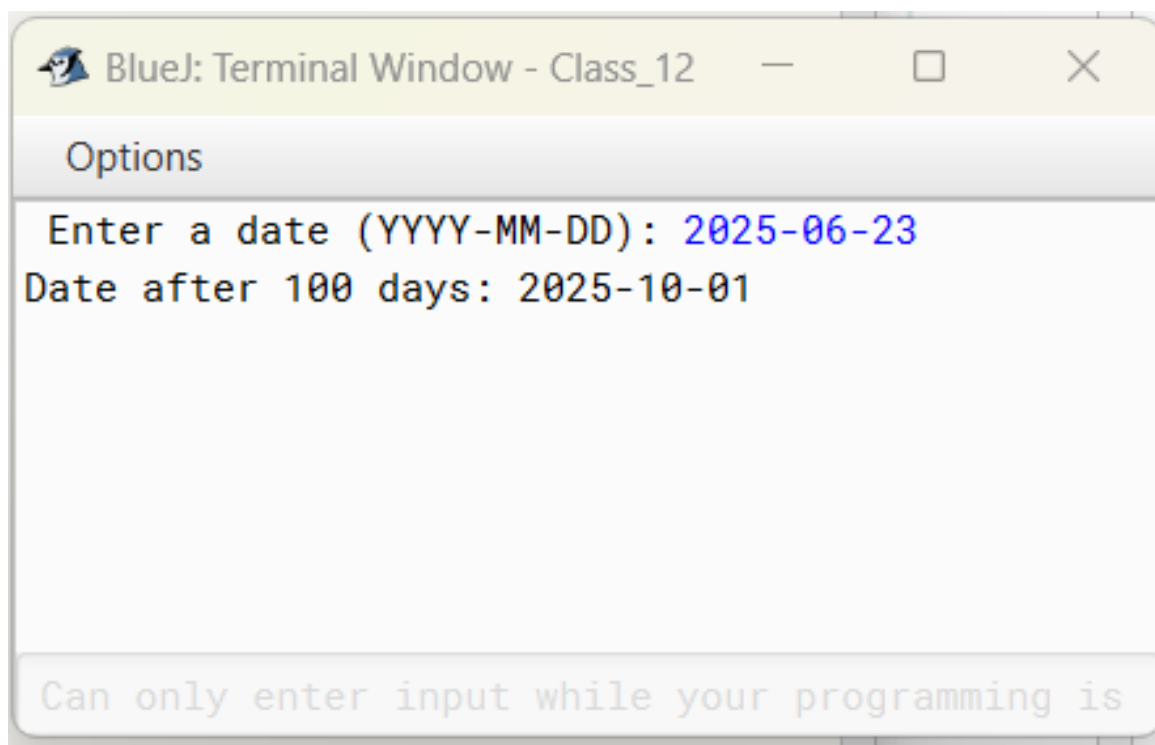
// Helper functions
public static boolean isLeapYear(int year) {
    return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
}

public static int daysInMonth(int year, int month) {
    int[] days = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
    if (month == 2 && isLeapYear(year)) {
        return 29;
    }
    return days[month];
}

public static int daysInYear(int year) {
    return isLeapYear(year) ? 366 : 365;
}
}
```

Class compiled - no syntax errors

saved



Num\_Dude - Class\_12

Class Edit Tools Options

Num\_Dude X

Compile Undo Cut Copy Paste Find... Close Source Code

```
import java.util.Scanner;
public class Num_Dude{
    int num;
    Num_Dude()
    {
        num=0;
    }
    void input()
    {
        Scanner sc = new Scanner(System.in);
        System.out.println("enter the number");
        num = sc.nextInt();
    }
    int sumDigits(int x)
    {
        if(x==0)
            return 0;
        else
            return x%10 +sumDigits(x/10);
    }
    void isDude()
    {
        int sdig=sumDigits(num);
        if((sdig*sdig*sdig)==num)
            System.out.println("Dudeny number");
        else
            System.out.println(" Not a Dudeny number");
    }
    public static void main(String args[]){
        Num_Dude obj=new Num_Dude();
        obj.input();
        obj.isDude();
    }
}
```

Class compiled - no syntax errors saved

BlueJ: Terminal Window - Class\_12

Options

enter the number

123

Not a Dudeny number

Can only enter input while your programming is

BlueJ: Terminal Window - Class\_12

Options

enter the number

512

Dudeny number

Can only enter input while your programming is

```
import java.io.*;
import java.util.Scanner;
class EqMat{
private int a[][];
private static int m;
private static int n;
public EqMat(int mm, int nn) {
m = mm;
n = nn;
a = new int[m][n];
}
public void readArray( )throws IOException {
Scanner sc = new Scanner(System.in);
for(int i = 0; i < m; i++) {
for(int j = 0; j < n; j++) {
a[i][j] = sc.nextInt();
}
}
}
public static boolean check(EqMat p, EqMat q) {
boolean flag = true;
for(int i = 0; i < m; i++) {
for(int j = 0; j < n; j++) {
if(p.a[i][j] !=q.a[i][j])
return false;
}
}
return flag;
}
public void print() {
for(int i = 0; i < m; i++) {
for(int j = 0; j < n; j++) {
System.out.print(a[i] [j] + " ");
}
System.out.println();
}
}

public static void main(String args[ ]) throws IOException {
Scanner sc = new Scanner(System.in);
System.out.println("Number of rows ");
int rows = sc.nextInt();
System.out.print("Number of columns ");
int columns = sc.nextInt();
EqMat obj1 = new EqMat(rows, columns);
EqMat obj2 = new EqMat(rows, columns);
System.out.println("Enter elements offirst matrix");
obj1.readArray();
System.out.println("Enter elements ofsecond matrix");
obj2.readArray();
System.out.println("First Matrix:");
obj1.print();
System.out.println("Second Matrix:");
obj2.print();
if(check(obj1, obj2))
{
System.out.println("Both Matrices are Equal");
}
else
{
System.out.println("Matrices are not Equal");
}
}
}
```

```
BlueJ: Terminal Window - Class_12
Options
Number of rows
4
Number of columns 4
Enter elements of first matrix
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Enter elements of second matrix
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
First Matrix:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Second Matrix:
Second Matrix:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Both Matrices are Equal
Can only enter input while your programming is
```



GoldbachNumber - Class\_12

Class Edit Tools Options

GoldbachNumber X

Compile Undo Cut Copy Paste Find... Close Source Code

```
import java.util.Scanner;

public class GoldbachNumber {

    public static boolean isPrime(int num) {
        if (num <= 1) {
            return false;
        }
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }

    public static void findOddPrimePairs(int number) {
        if (number <= 2 || number % 2 != 0) {
            System.out.println("Invalid input: The number must be an even integer greater than 2.");
            return;
        }

        System.out.println("Odd prime pairs for " + number + ":");
        for (int i = 3; i <= number / 2; i += 2) {
            if (isPrime(i) && isPrime(number - i)) {
                System.out.println(i + " and " + (number - i));
            }
        }
    }

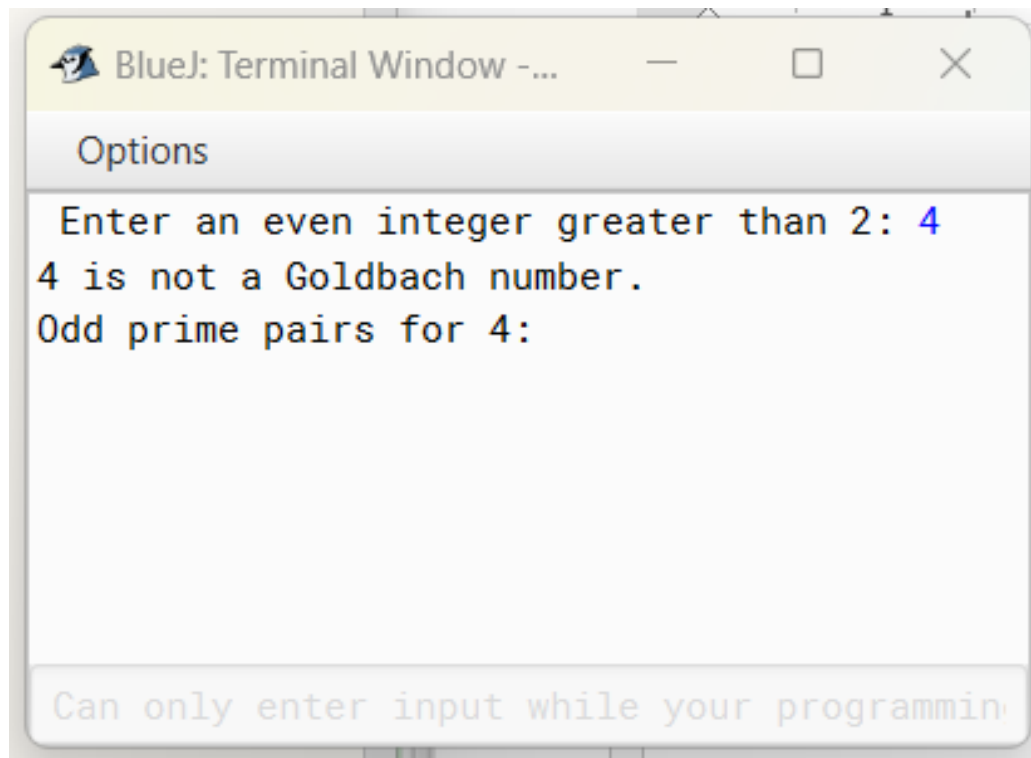
    public static boolean isGoldbachNumber(int number) {
        if (number <= 2 || number % 2 != 0) {
            return false;
        }
        for (int i = 3; i <= number / 2; i += 2) {
            if (isPrime(i) && isPrime(number - i)) {
                return true;
            }
        }
        return false;
    }

    public static void main(String[] args) {
        GoldbachNumber obj = new GoldbachNumber();
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter an even integer greater than 2: ");
        int n = scanner.nextInt();

        if (isGoldbachNumber(n)) {
            System.out.println(n + " is a Goldbach number.");
        } else {
            System.out.println(n + " is not a Goldbach number.");
        }

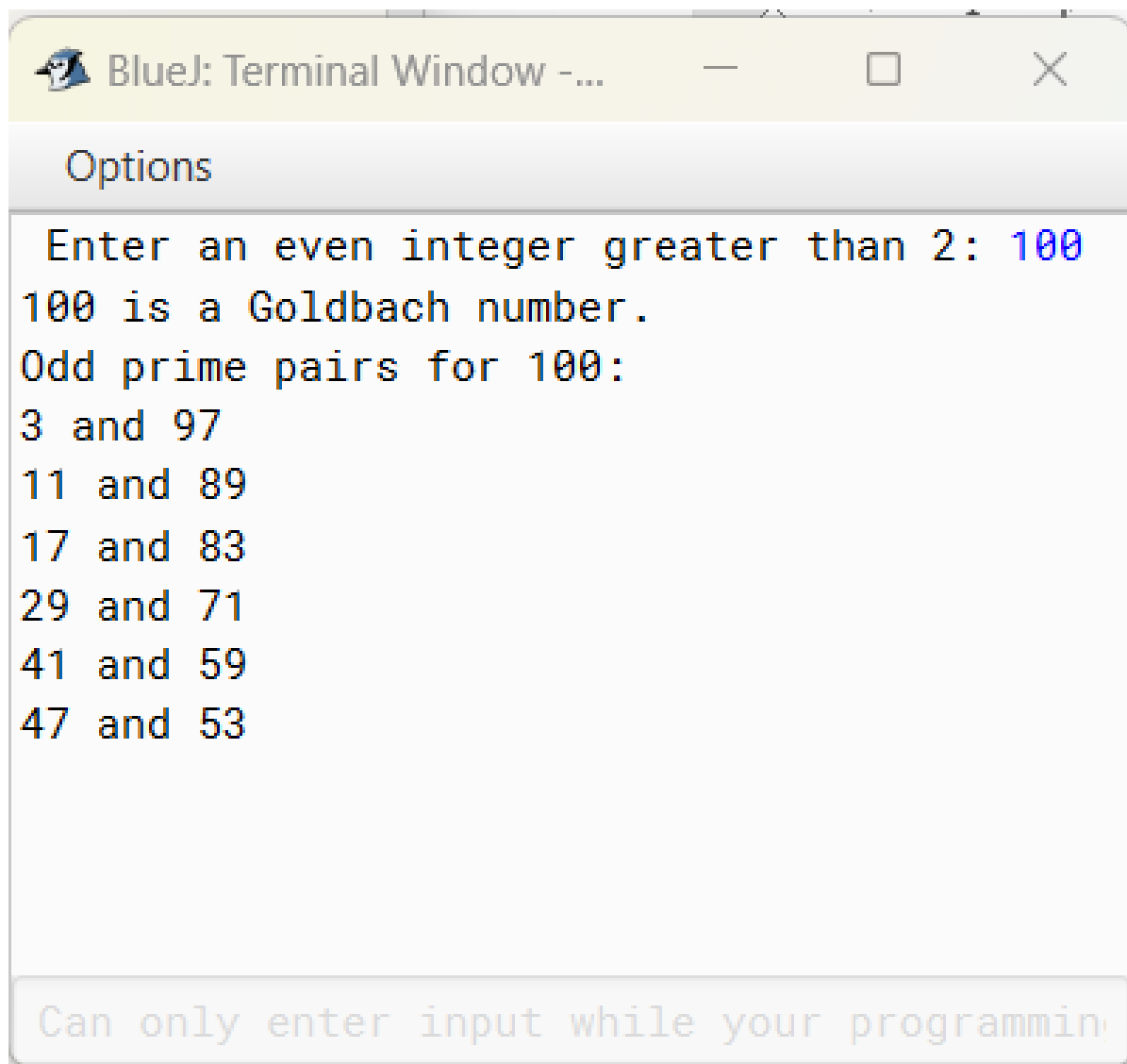
        obj.findOddPrimePairs(n);
        scanner.close();
    }
}
```

Class compiled - no syntax errors saved



A terminal window titled "BlueJ: Terminal Window -..." with standard window controls. It contains a section titled "Options" with the following text: "Enter an even integer greater than 2: 4", "4 is not a Goldbach number.", and "Odd prime pairs for 4:". At the bottom, a greyed-out message reads "Can only enter input while your programmin".

```
BlueJ: Terminal Window -...  
Options  
Enter an even integer greater than 2: 4  
4 is not a Goldbach number.  
Odd prime pairs for 4:  
  
Can only enter input while your programmin
```



A terminal window titled "BlueJ: Terminal Window -..." with standard window controls. It contains a section titled "Options" with the following text: "Enter an even integer greater than 2: 100", "100 is a Goldbach number.", "Odd prime pairs for 100:", and a list of prime pairs: "3 and 97", "11 and 89", "17 and 83", "29 and 71", "41 and 59", and "47 and 53". At the bottom, a greyed-out message reads "Can only enter input while your programmin".

```
BlueJ: Terminal Window -...  
Options  
Enter an even integer greater than 2: 100  
100 is a Goldbach number.  
Odd prime pairs for 100:  
3 and 97  
11 and 89  
17 and 83  
29 and 71  
41 and 59  
47 and 53  
  
Can only enter input while your programmin
```

```
import java.util.Scanner;

public class LargestElement2DArray {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int rows = scanner.nextInt();
        System.out.print("Enter the number of columns: ");
        int columns = scanner.nextInt();

        int[][] array = new int[rows][columns];

        System.out.println("Enter the elements of the array:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                array[i][j] = scanner.nextInt();
            }
        }

        int largestElement = array[0][0];
        int largestElementRow = 0;
        int largestElementColumn = 0;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                if (array[i][j] > largestElement) {
                    largestElement = array[i][j];
                    largestElementRow = i;
                    largestElementColumn = j;
                }
            }
        }

        System.out.println("Largest element: " + largestElement);
        System.out.println("Position: Row " + largestElementRow + ", Column " + largestElementColumn);

        int[] temp = array[0];
        array[0] = array[largestElementRow];
        array[largestElementRow] = temp;

        System.out.println("Array after moving largest element's row to the first row:");
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                System.out.print(array[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

## Options

Enter the number of rows: 4

Enter the number of columns: 4

Enter the elements of the array:

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

Largest element: 16

Position: Row 3, Column 3

Array after moving largest element's row to the first row:

13 14 15 16

5 6 7 8

9 10 11 12

1 2 3 4

Can only enter input while your programming is running

LCM - Class\_12

Class Edit Tools Options

LargestElement2DArray X LCM X

Compile Undo Cut Copy Paste Find... Close Source Code

```
import java.util.*;
class LCM
{
    int n1,n2;
    int large,sm;
    int l;
    static Scanner sc=new Scanner(System.in);
    void accept()
    {
        System.out.println("enter 2 different integers:");
        n1=sc.nextInt();
        n2=sc.nextInt();
        if (n1>n2)
        {
            large=n1;
            sm=n2;
        }
        else if (n2>n1);
        { large=n2;
            sm=n1;
        }
    }
    int getLCM()
    { if(large!=sm)
        {
            if (large>sm)
                large=large-sm;
            else if (large<sm)
                sm=sm-large;
            return getLCM();
        }
        else
            return (n1*n2)/large;
    }
    void display()
    {
        l=getLCM();
        System.out.println("LCM of "+n1 +"and "+n2+"="+l);
    }
    public static void main()
    {
        LCM ob=new LCM();
        ob.accept();
        ob.display();
    }
}
```

Class compiled - no syntax errors saved

BlueJ: Terminal Window - Class\_12

Options

enter 2 different integers:

4

6

LCM of 4and 6=12

Can only enter input while your programming is runni

```
import java.util.Scanner;
public class MatSymm{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number of rows and coloumns");
        int M = sc.nextInt();
        int arr[][] = new int[M][M];
        int ld =0, rd =0;
        if(M>2&&M<10)
        {
            System.out.println("Enter the elements");
            for(int i=0;i<M;i++)
            {
                for(int j =0;j<M;j++)
                {
                    arr[i][j]= sc.nextInt();
                }
            }
            //Checking if the matrix is symmetric or not
            for(int i=0;i<M;i++)
            {
                for(int j =0;j<M;j++)
                {
                    if(arr[i][j]== arr[j][i])
                        System.out.println("The Given Matrix is Symmetric");
                }
            }
            //Calculating the sum of left diagonal and right diagonal elements
            for(int i=0;i<M;i++)
            {
                for(int j =0;j<M;j++)
                {
                    if(i==j)
                        ld+= arr[i][j];
                }
            }
            System.out.println("Sum of right diagonal =" +rd);
            //Original Matrix
            for(int i=0;i<M;i++)
            {
                for(int j =0;j<M;j++)
                {
                    System.out.print(arr[i][j]+" ");
                }
                System.out.println();
            }
        }
    }
}
```

## Options

Enter the number of rows and coloumns

3

Enter the elements

3

1

2

3

4

5

6

7

8

The Given Matrix is Symmeteric

The Given Matrix is Symmeteric

The Given Matrix is Symmeteric

Sum of left diagonal =15

Sum of right diagonal =12

3 1 2

3 4 5

6 7 8

Can only enter input while your programming is runni

























